

Base rule to compute partial derivatives $\leadsto$ Chain rule:

$$\boxed{\frac{\partial e(\mathbf{y}, \hat{\mathbf{y}})}{\partial \omega_{ij}^{(l)}} = \frac{\partial e}{\partial z_i^{(l)}} \frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}}} \quad (1)$$

Base rule to compute partial derivatives $\leadsto$ Chain rule:

$$\boxed{\frac{\partial e(\mathbf{y}, \hat{\mathbf{y}})}{\partial \omega_{ij}^{(l)}} = \frac{\partial e}{\partial z_i^{(l)}} \frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}}} \quad (1)$$

$$\frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}} = \frac{\partial (\sum_k \omega_{ik}^{(l)} a_k^{(l-1)} + b_i^{(l-1)})}{\partial \omega_{ij}^{(l)}} = a_j^{(l-1)}$$

Base rule to compute partial derivatives $\leadsto$ Chain rule:

$$\boxed{\frac{\partial e(\mathbf{y}, \hat{\mathbf{y}})}{\partial \omega_{ij}^{(l)}} = \frac{\partial e}{\partial z_i^{(l)}} \frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}}} \quad (1)$$

$$\frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}} = \frac{\partial (\sum_k \omega_{ik}^{(l)} a_k^{(l-1)} + b_i^{(l-1)})}{\partial \omega_{ij}^{(l)}} = a_j^{(l-1)}$$

let $\frac{\partial e}{\partial z_i^{(l)}}$ be the **delta term** noted $\delta_i^{(l)}$

Base rule to compute partial derivatives $\leadsto$ Chain rule:

$$\boxed{\frac{\partial e(\mathbf{y}, \hat{\mathbf{y}})}{\partial \omega_{ij}^{(l)}} = \frac{\partial e}{\partial z_i^{(l)}} \frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}}} \quad (1)$$

$$\frac{\partial z_i^{(l)}}{\partial \omega_{ij}^{(l)}} = \frac{\partial (\sum_k \omega_{ik}^{(l)} a_k^{(l-1)} + b_i^{(l-1)})}{\partial \omega_{ij}^{(l)}} = a_j^{(l-1)}$$

let $\frac{\partial e}{\partial z_i^{(l)}}$ be the **delta term** noted $\delta_i^{(l)}$

$$\boxed{\frac{\partial e}{\partial \omega_{ij}^{(l)}} = \delta_i^{(l)} a_j^{(l-1)}}$$

- e depends on pre-activation $z_i^{(l)}$ through all the pre-activations $z_j^{(l+1)}$ in layer $(l + 1)$ connected to the activation $a_i^{(l)}$
- let $N_i^{(l+1)} = \{ j \text{ in layer } (l + 1) \mid z_j^{(l+1)} \text{ is connected to } a_i^{(l)} \}$

- e depends on pre-activation $z_i^{(l)}$ through all the pre-activations $z_j^{(l+1)}$ in layer $(l+1)$ connected to the activation $a_i^{(l)}$
- let $N_i^{(l+1)} = \{ j \text{ in layer } (l+1) \mid z_j^{(l+1)} \text{ is connected to } a_i^{(l)} \}$

Once again thanks to **Chain rule**:

$$\delta_i^{(l)} = \frac{\partial e}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \frac{\partial e}{\partial z_j^{(l+1)}} \frac{\partial z_j^{(l+1)}}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} \frac{\partial a_i^{(l)}}{\partial z_j^{(l)}}$$

- e depends on pre-activation $z_i^{(l)}$ through all the pre-activations $z_j^{(l+1)}$ in layer $(l+1)$ connected to the activation $a_i^{(l)}$
- let $N_i^{(l+1)} = \{ j \text{ in layer } (l+1) \mid z_j^{(l+1)} \text{ is connected to } a_i^{(l)} \}$

Once again thanks to **Chain rule**:

$$\delta_i^{(l)} = \frac{\partial e}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \frac{\partial e}{\partial z_j^{(l+1)}} \frac{\partial z_j^{(l+1)}}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} \frac{\partial a_i^{(l)}}{\partial z_j^{(l)}}$$

given $\frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} = \frac{\partial (\sum_k \omega_{jk}^{(l+1)} a_k^{(l)} + b_l^{(l+1)})}{\partial a_i^{(l)}} = \omega_{ji}^{(l+1)}$ we get

- e depends on pre-activation $z_i^{(l)}$ through all the pre-activations $z_j^{(l+1)}$ in layer $(l+1)$ connected to the activation $a_i^{(l)}$
- let $N_i^{(l+1)} = \{ j \text{ in layer } (l+1) \mid z_j^{(l+1)} \text{ is connected to } a_i^{(l)} \}$

Once again thanks to **Chain rule**:

$$\delta_i^{(l)} = \frac{\partial e}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \frac{\partial e}{\partial z_j^{(l+1)}} \frac{\partial z_j^{(l+1)}}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} \frac{\partial a_i^{(l)}}{\partial z_j^{(l)}}$$

given $\frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} = \frac{\partial (\sum_k \omega_{jk}^{(l+1)} a_k^{(l)} + b_j^{(l+1)})}{\partial a_i^{(l)}} = \omega_{ji}^{(l+1)}$ we get

$$\delta_i^{(l)} = \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \omega_{ji}^{(l+1)} \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} = \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \omega_{ji}^{(l+1)}$$

- e depends on pre-activation $z_i^{(l)}$ through all the pre-activations $z_j^{(l+1)}$ in layer $(l+1)$ connected to the activation $a_i^{(l)}$
- let $N_i^{(l+1)} = \{ j \text{ in layer } (l+1) \mid z_j^{(l+1)} \text{ is connected to } a_i^{(l)} \}$

Once again thanks to **Chain rule**:

$$\delta_i^{(l)} = \frac{\partial e}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \frac{\partial e}{\partial z_j^{(l+1)}} \frac{\partial z_j^{(l+1)}}{\partial z_i^{(l)}} = \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} \frac{\partial a_i^{(l)}}{\partial z_j^{(l)}}$$

given $\frac{\partial z_j^{(l+1)}}{\partial a_i^{(l)}} = \frac{\partial (\sum_k \omega_{jk}^{(l+1)} a_k^{(l)} + b_j^{(l+1)})}{\partial a_i^{(l)}} = \omega_{ji}^{(l+1)}$ we get

$$\delta_i^{(l)} = \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \omega_{ji}^{(l+1)} \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} = \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \omega_{ji}^{(l+1)}$$

Delta rule $\delta_i^{(l)} = f'^{(l)}(z_i^{(l)}) \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \omega_{ji}^{(l+1)}$ $f^{(l)}$: layer (l) activ. func.